

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 19

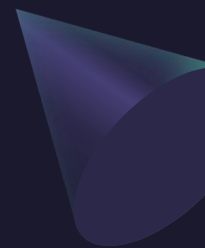


Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 13.1-13.3 (vector functions, their integrals and derivatives, and arc length)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



ALVARO: Start the recording!

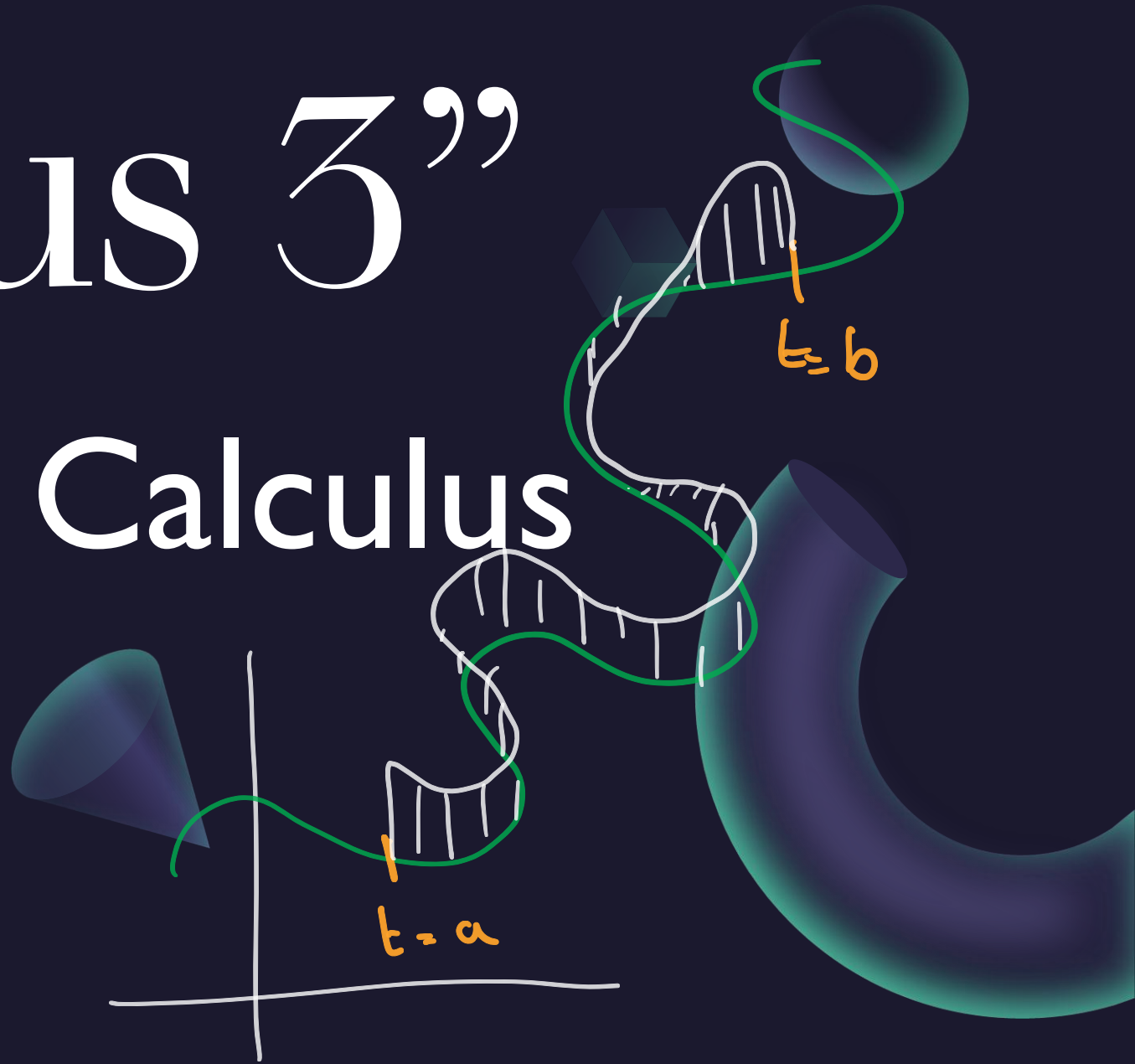


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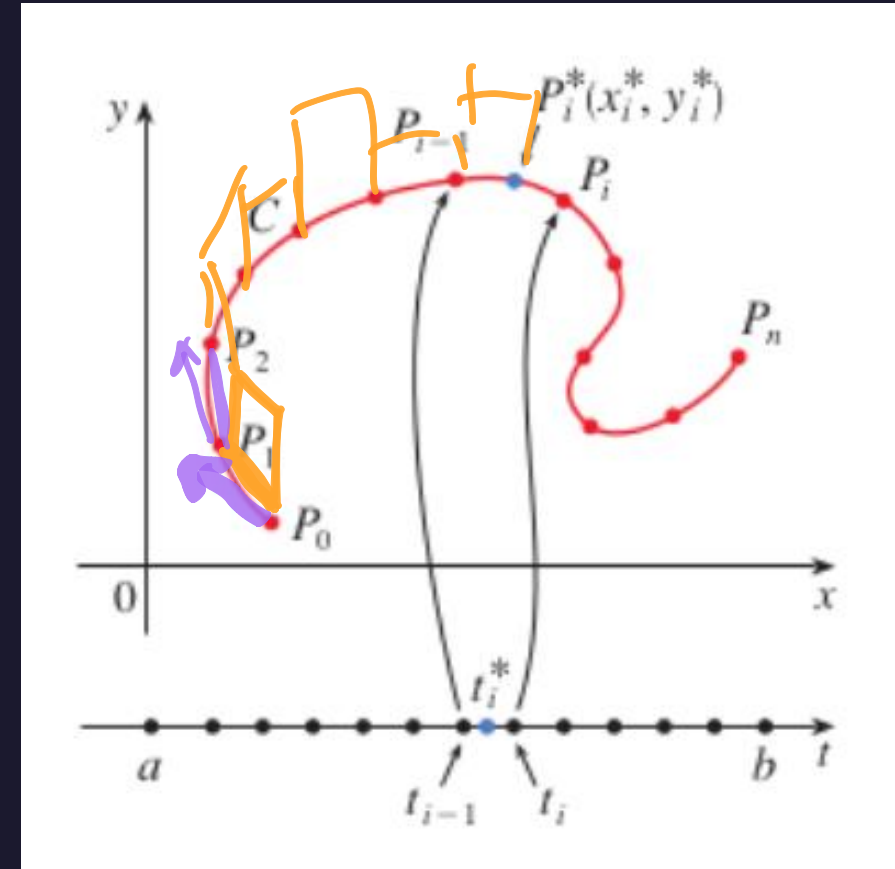
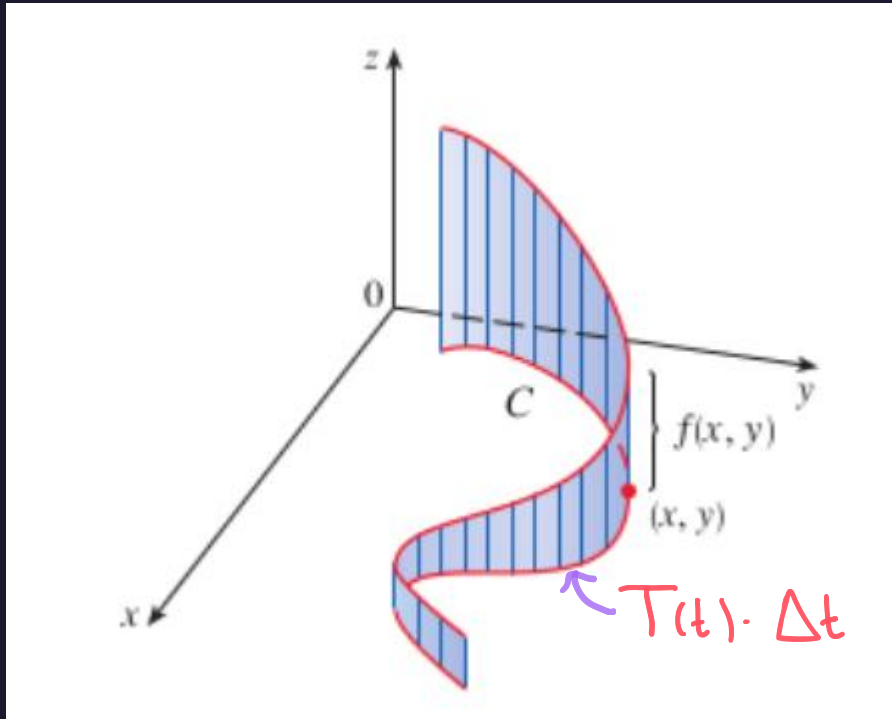
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Line Integrals

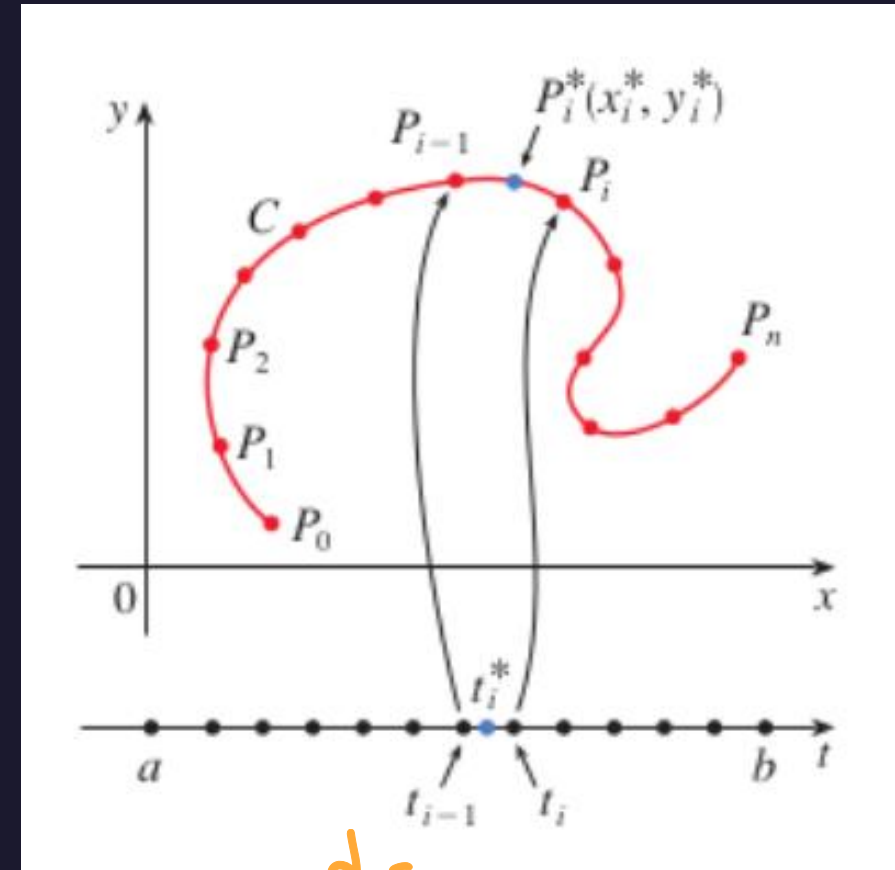
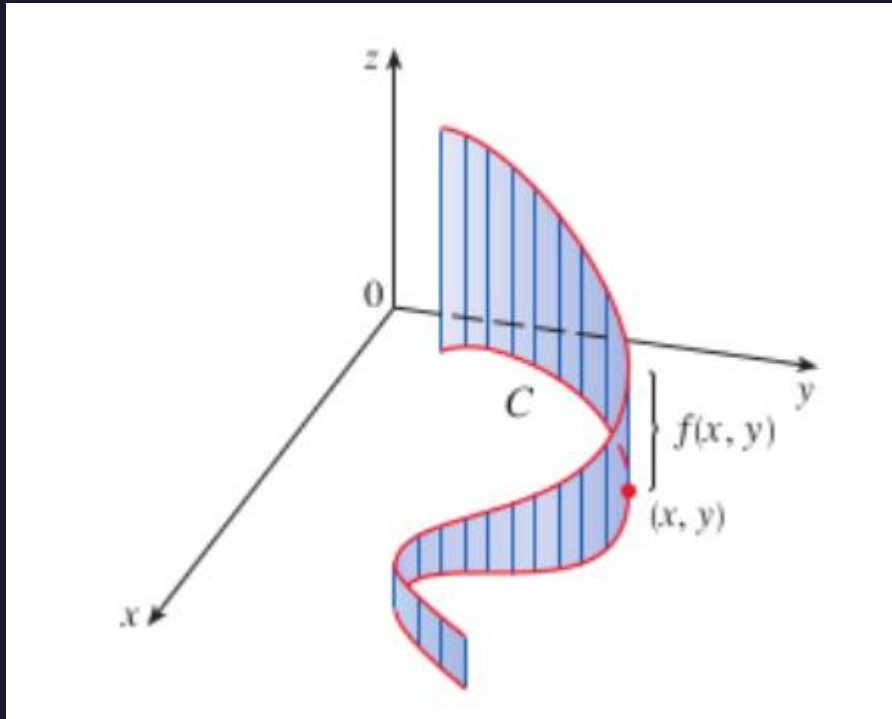


Line Integrals



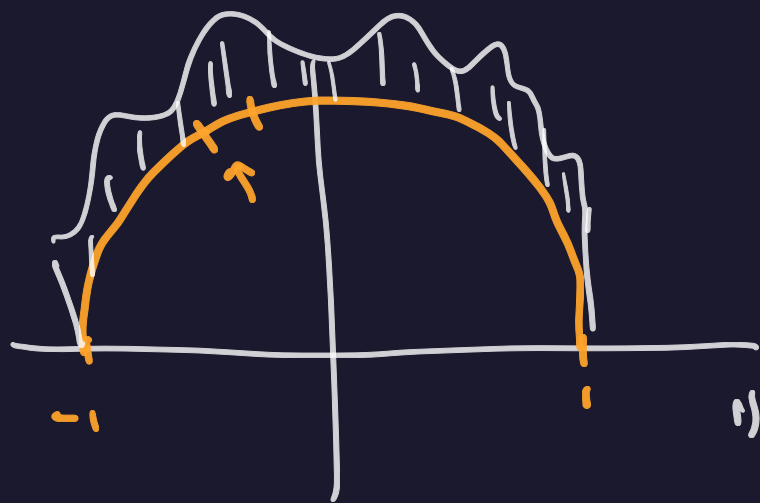
$$\int_C f(x, y) ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i$$

Line Integrals



$$\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Example: Evaluate $\int_C (2 + x^2 y) ds$ where C is the upper half of the unit circle.



1) PARAMETRIZE

$$r(t) = (\cos t, \sin t)$$

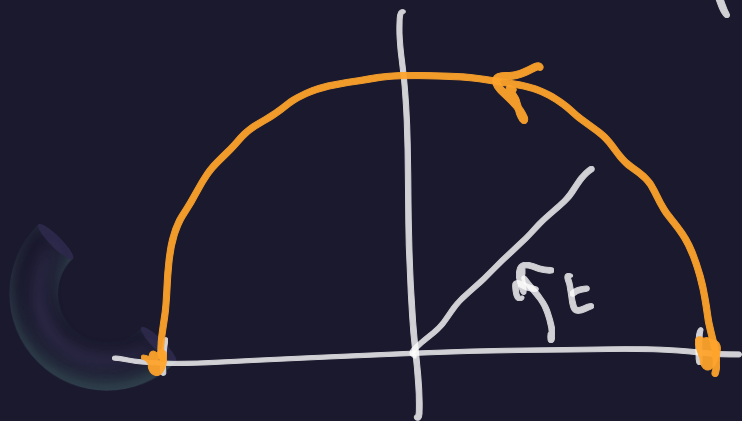
$$0 \leq t \leq \pi$$

2) TANGENT VECTOR

$$r'(t) = (-\sin t, \cos t)$$

3) $|r'(t)|$

$$\begin{aligned} &= \sqrt{(-\sin t)^2 + (\cos t)^2} \\ &= \sqrt{\sin^2 t + \cos^2 t} = \sqrt{1} = 1 \end{aligned}$$



$$r(t) = (x(t), y(t)) = (\cos t, \sin t)$$

Example: Evaluate $\int_C (2 + x^2 y) ds$ where C is the upper half of the unit circle.

$$\int_C (2 + x^2 y) ds = \int_0^\pi (2 + \cos^2 t \cdot \sin t) \cdot \overset{|r'(t)|}{1} \cdot dt$$

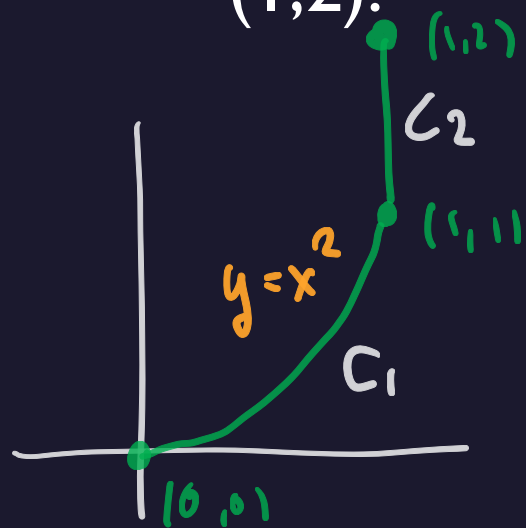
$$= \int_0^\pi 2 dt + \int_0^\pi \cos^2 t \cdot \sin t dt$$

$u = \cos t \rightarrow \begin{matrix} t=0 & \cos 0 = 1 \\ t=\pi & \cos \pi = -1 \end{matrix}$
 $du = -\sin t dt$

$$= 2\pi - \int_1^{-1} u^2 du = 2\pi + \underbrace{\frac{u^3}{3}}_{2/3} \Big|_{-1}^1 = \boxed{2\pi + 2/3}$$

$$= \int_{-1}^1 u^2 du$$

Example: Evaluate $\int_C (2x) ds$ where C consists of the arc of the parabola $y = x^2$ from $(0,0)$ to $(1,1)$ followed by a line segment from $(1,1)$ to $(1,2)$.



$$\int_C (2x) ds = \int_{C_1} (2x) ds + \int_{C_2} (2x) ds$$

$$C_1 = \{(t, t^2) \mid 0 \leq t \leq 1\}$$

$$C_2 = \{(1, t) \mid 1 \leq t \leq 2\}$$

$$r_1(t) = (t, t^2)$$

$$r'_1(t) = (1, 2t)$$

$$|r'_1(t)| = \sqrt{1^2 + 4t^2} = \sqrt{1 + 4t^2}$$

$$r_2(t) = (1, t)$$

$$r'_2(t) = (0, 1)$$

$$|r'_2(t)| = \sqrt{0^2 + 1^2} = 1$$

Example: Evaluate $\int_C (2x) ds$ where C consists of the arc of the parabola $y = x^2$ from $(0,0)$ to $(1,1)$ followed by a line segment from $(1,1)$ to $(1,2)$.

$$\begin{aligned} \int_{C_1} 2x ds &= \int_0^1 2t \cdot \sqrt{1+4t^2} dt = du = 8t dt \\ &= \int_1^5 \frac{1}{4} (u)^{1/2} du = \frac{1}{4} \frac{u^{3/2}}{3/2} \Big|_1^5 = \boxed{\frac{5\sqrt{5}-1}{6}} \end{aligned}$$

$u = 1+4t^2 \begin{cases} t=0 & u=1 \\ t=1 & u=5 \end{cases}$

$$\int_{C_2} (2x) ds = \int_1^2 2 \cdot 1 \cdot 1 dt = \boxed{2}$$

$$\int_C 2x ds = \boxed{\frac{5\sqrt{5}-1}{6} + 2}$$

Application of Line Integrals: Center of Mass

$$(\bar{x}, \bar{y})$$

$$\bar{x} = \frac{1}{m} \int_C x \rho(x, y) ds \qquad \bar{y} = \frac{1}{m} \int_C y \rho(x, y) ds$$

Example: A wire takes the shape of the top half of a semi-circle of radius 1 and it is thicker near its base than near the top. Find the center of mass of the wire if the linear density at any point is proportional to its distance from the line $y = 1$.

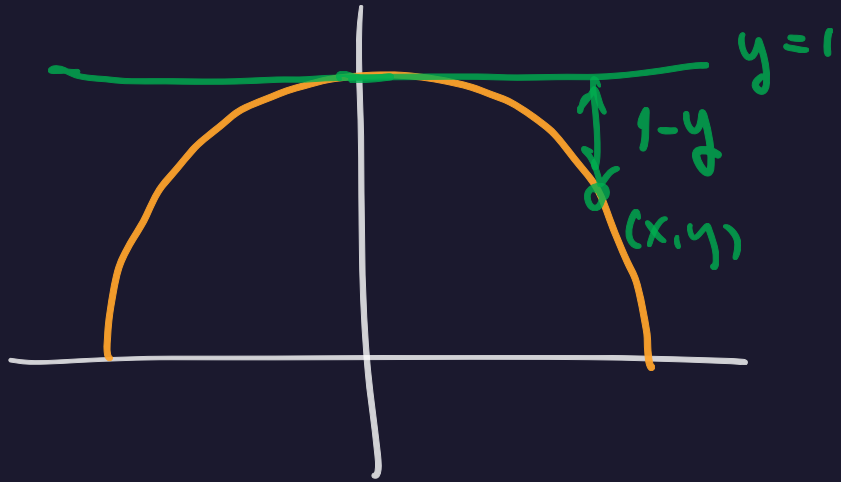
$$r(t) = (\cos(t), \sin(t)) \quad 0 \leq t \leq \pi$$

$$\rho(x, y) = k(1 - y) \quad \text{density}$$

$$M = \int_C \rho(x, y) \, ds$$

$$M = \boxed{k(\pi - 2)}$$

$$\int_0^\pi \rho(x(t), y(t)) \cdot |r'(t)| \, dt = \int_0^\pi k(1 - \sin(t)) \cdot 1 \, dt$$



Example: A wire takes the shape of the top half of a semi-circle of radius 1 and it is thicker near its base than near the top. Find the center of mass of the wire if the linear density at any point is proportional to its distance from the line $y = 1$.

$$\bar{x} = \frac{1}{k(\pi-2)} \int_0^{\pi} \cos t \cdot k(1-\sin t) dt = 0$$

$$\sin^2 t = \frac{1 - \cos(2t)}{2}$$

$$\bar{y} = \frac{1}{k(\pi-2)} \int_0^{\pi} \sin t \cdot k(1-\sin t) dt = \frac{1}{\pi-2} \int_0^{\pi} \sin t - \sin^2 t dt$$

$= \dots =$

$$\frac{4-\pi}{2(\pi-2)}$$

Center of Mass = $\left(0, \frac{4-\pi}{2(\pi-2)}\right)$

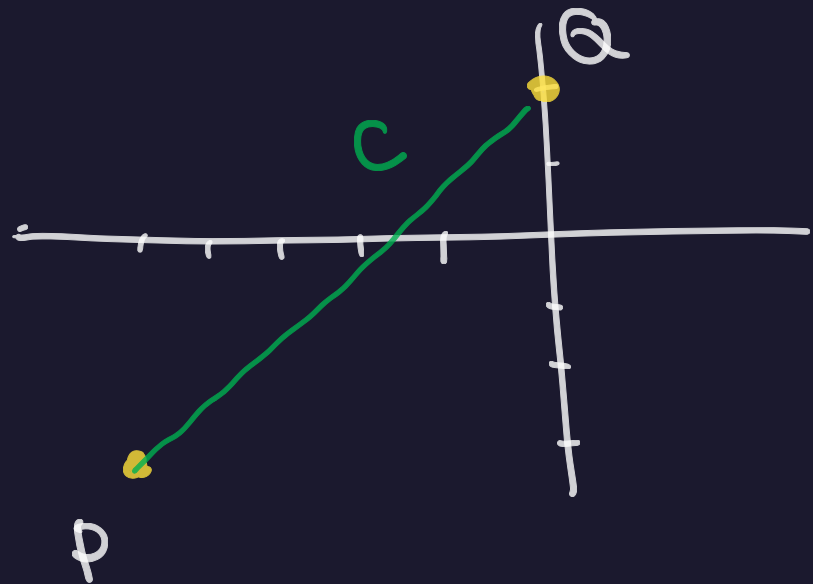
Line integrals with respect to x and y $\int_C f(x, y) ds$

$$\begin{aligned}\int_C f(x, y) dx &= \int_a^b f(x(t), y(t)) x'(t) dt \\ \int_C f(x, y) dy &= \int_a^b f(x(t), y(t)) y'(t) dt\end{aligned}$$

$$\int_C f(x(t), y(t)) \sqrt{x'^2 + y'^2} dt$$

$$\int_C P(x, y) dx + \int_C Q(x, y) dy = \int_C P(x, y) dx + Q(x, y) dy$$

Example: Evaluate $\int_C y^2 dx + x dy$ where C is the line segment from $(-5, -3)$ to $(0, 2)$.



$$r(t) = (1-t)P + tQ \quad 0 \leq t \leq 1$$

$$= (1-t)(-5, -3) + t(0, 2)$$

$$= (-5 + 5t, -3 + 3t + 2t)$$

$$= (-5 + 5t, -3 + 5t) \quad 0 \leq t \leq 1$$

$$r'(t) = (5, 5)$$

$$dx = x'(t) dt = 5 dt, \quad dy = y'(t) dt = 5 dt$$

$$r(t) = (-5 + 5t, -3 + 5t)$$

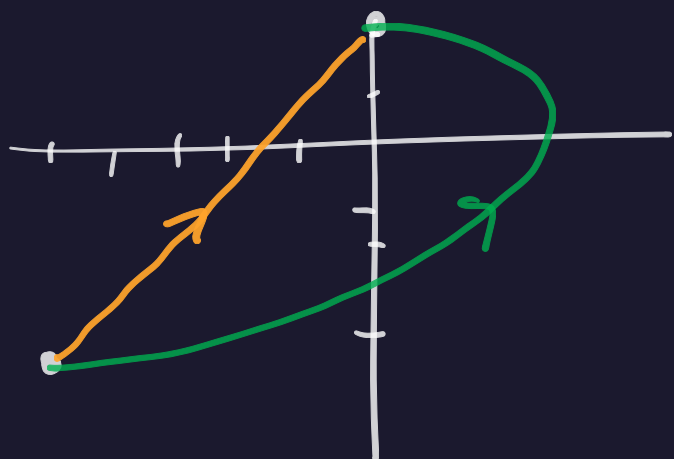
Example: Evaluate $\int_C y^2 dx + x dy$ where C is the line segment from $(-5, -3)$ to $(0, 2)$.

$$\int_C y^2 dx + x dy = \int_0^1 ((-3 + 5t)^2 \cdot 5 + (-5 + 5t) \cdot 5) dt$$

$$\dots = \int_0^1 (20 - 125t + 125t^2) dt$$

$$\dots = \boxed{-\frac{5}{6}}$$

Example: Evaluate $\int_C y^2 dx + x dy$ where C is the arc of the parabola $x = 4 - y^2$ from $(-5, -3)$ to $(0, 2)$.



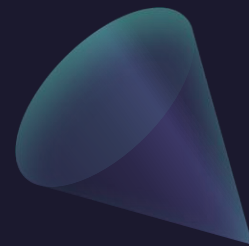
$$r(t) = (4 - t^2, t) \quad -3 \leq t \leq 2$$

$$r'(t) = (-2t, 1)$$

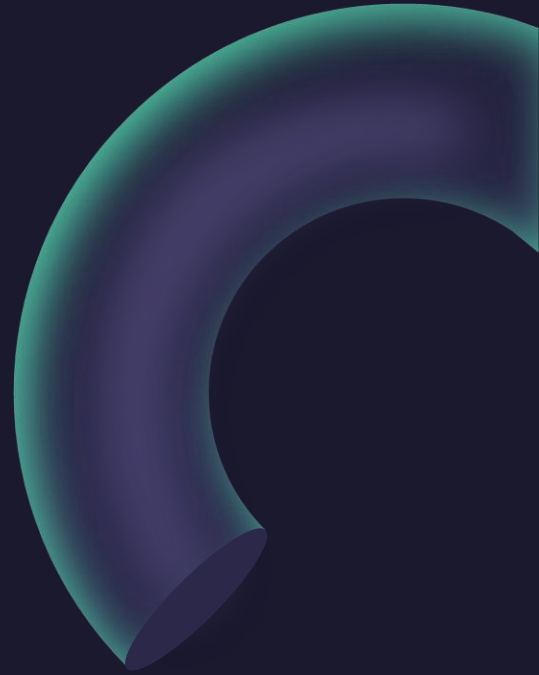
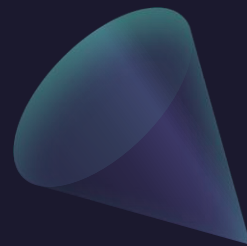
$$dx = -2t dt \quad dy = 1 dt = dt$$

$$\begin{aligned} \int_C y^2 dx + x dy &= \int_{-3}^2 (t^2(-2t) + (4 - t^2)) dt = \int_{-3}^2 (-2t^3 + 4 - t^2) dt \\ &= \dots = \boxed{\frac{245}{6}} \end{aligned}$$

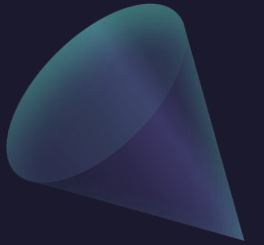
Example: Evaluate $\int_C y^2 dx + x dy$ where C is the line segment from $(-5,-3)$ to $(0,2)$.



Example:

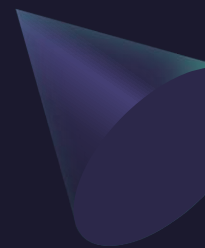


Questions?





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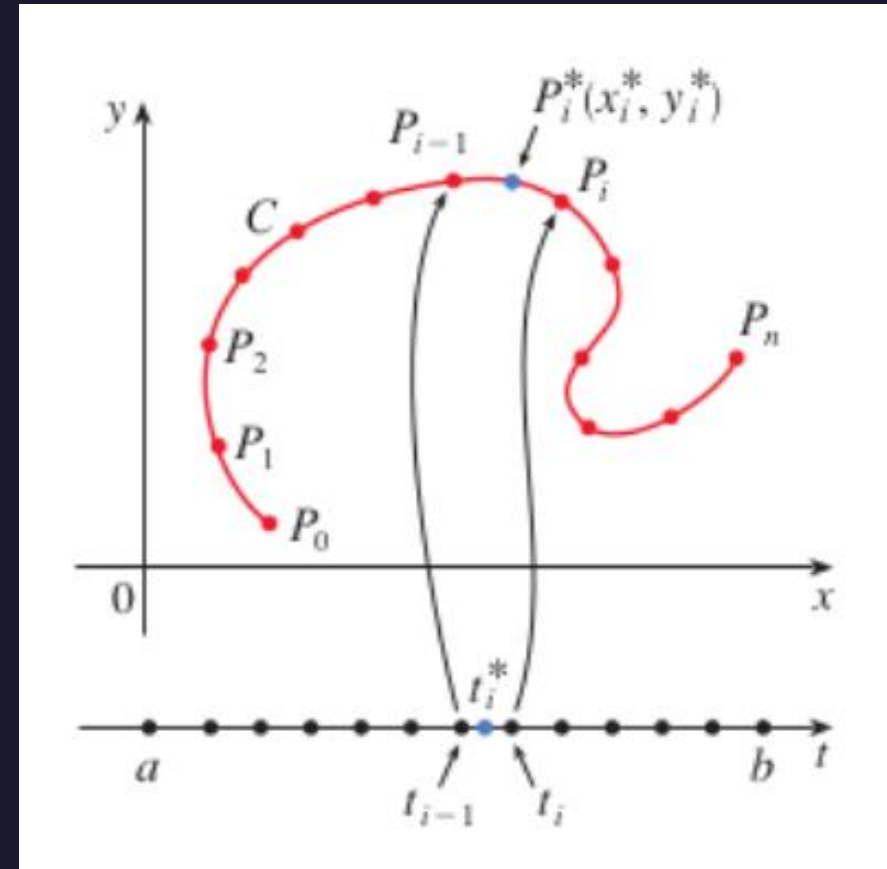
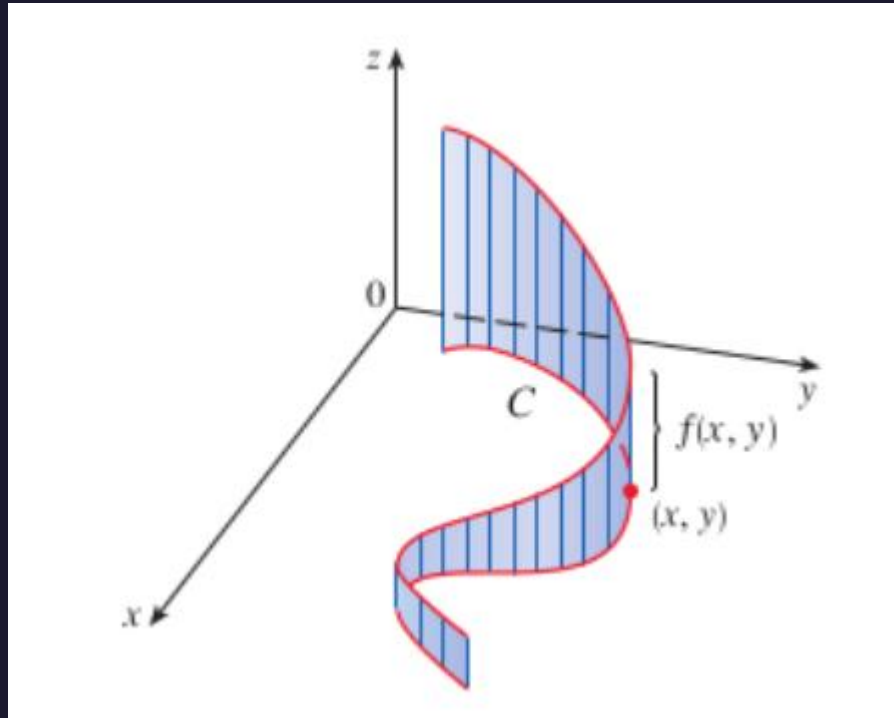
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Line Integrals... in Space



Line Integrals



$$\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

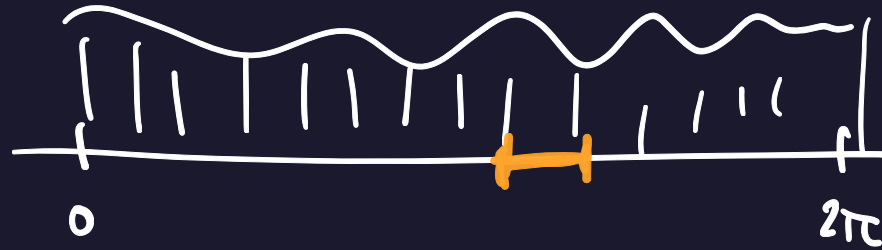
Line Integrals

$$\int_C f(x, y, z) \, ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \, dt$$

$$\int_C P(x, y, z) \, dx + Q(x, y, z) \, dy + R(x, y, z) \, dz$$

$$\int_a^b f(x(t), y(t), z(t)) \cdot \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \, dt$$

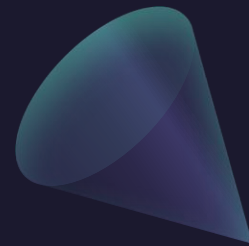
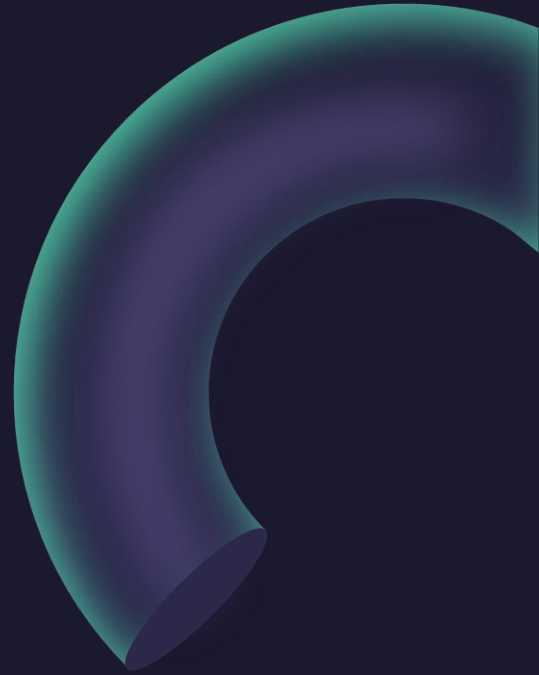
Example: Evaluate $\int_C y \sin(z) ds$ where C is the helix given by $\mathbf{r}(t) = (\cos(t), \sin(t), t)$, $0 \leq t \leq 2\pi$



$$\mathbf{r}'(t) = (-\sin t, \cos t, 1)$$

$$|\mathbf{r}'(t)| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{2}$$

$$ds = |\mathbf{r}'(t)| \cdot dt = \sqrt{2} dt$$

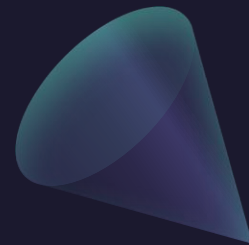
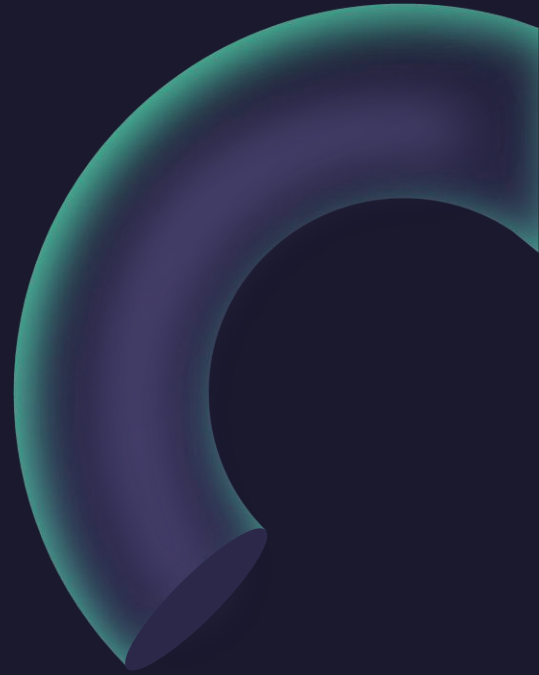


Example: Evaluate $\int_C y \sin(z) ds$ where C is the helix given by $(\cos(t), \sin(t), t)$, $0 \leq t \leq 2\pi$

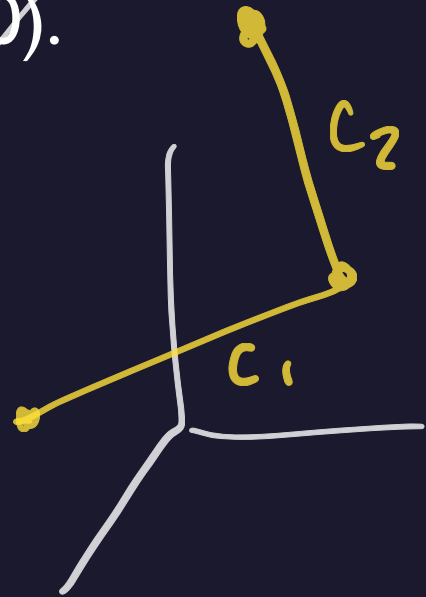
$$\int_C y \cdot \sin(z) ds = \int_0^{2\pi} \sin t \cdot \sin(t) \cdot \sqrt{2} dt$$

$$= \sqrt{2} \int_0^{2\pi} \sin^2 t dt = \dots = \boxed{\pi\sqrt{2}}$$

$$\searrow \sin^2 t = \frac{1 - \cos 2t}{2}$$



Example: Evaluate $\int_C y \, dx + z \, dy + x \, dz$ where C consists of the line segment from $(2,0,0)$ to $(3,4,5)$, followed by the line segment from $(3,4,5)$ to $(3,4,0)$.



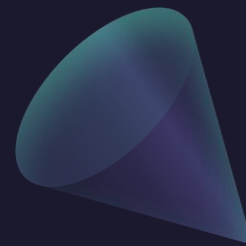
$$\int_C = \int_{C_1} + \int_{C_2}$$

$$\begin{aligned} r_1(t) &= (1-t)(2,0,0) + t(3,4,5) \quad 0 \leq t \leq 1 \\ &= (2-2t+3t, 4t, 5t) = (2+t, 4t, 5t) \end{aligned}$$

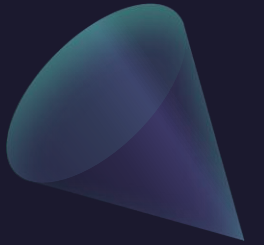
$$r'_1(t) = (1, 4, 5) \quad dx = 1dt \quad dy = 4dt \quad dz = 5dt$$

$$\int_0^1 (4t \cdot 1 + 5t \cdot 4 + (2+t) \cdot 5) dt = \int_{C_1}$$

Example: Evaluate $\int_C y \, dx + z \, dy + x \, dz$ where C consists of the line segment from $(2,0,0)$ to $(3,4,5)$, followed by the line segment from $(3,4,5)$ to $(3,4,0)$.

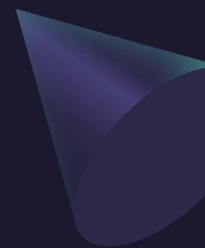


Questions?





ALVARO: Start the recording!

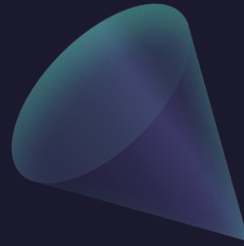
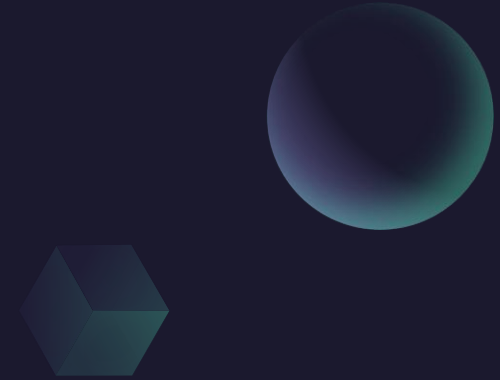


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Multi-Variable Calculus

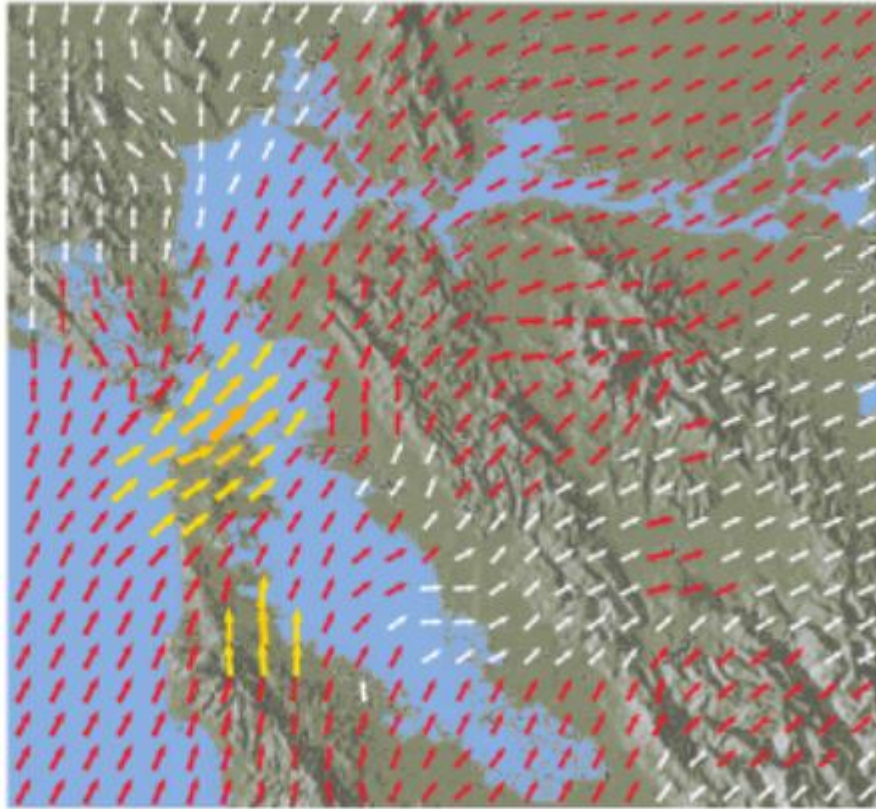
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Vector Fields

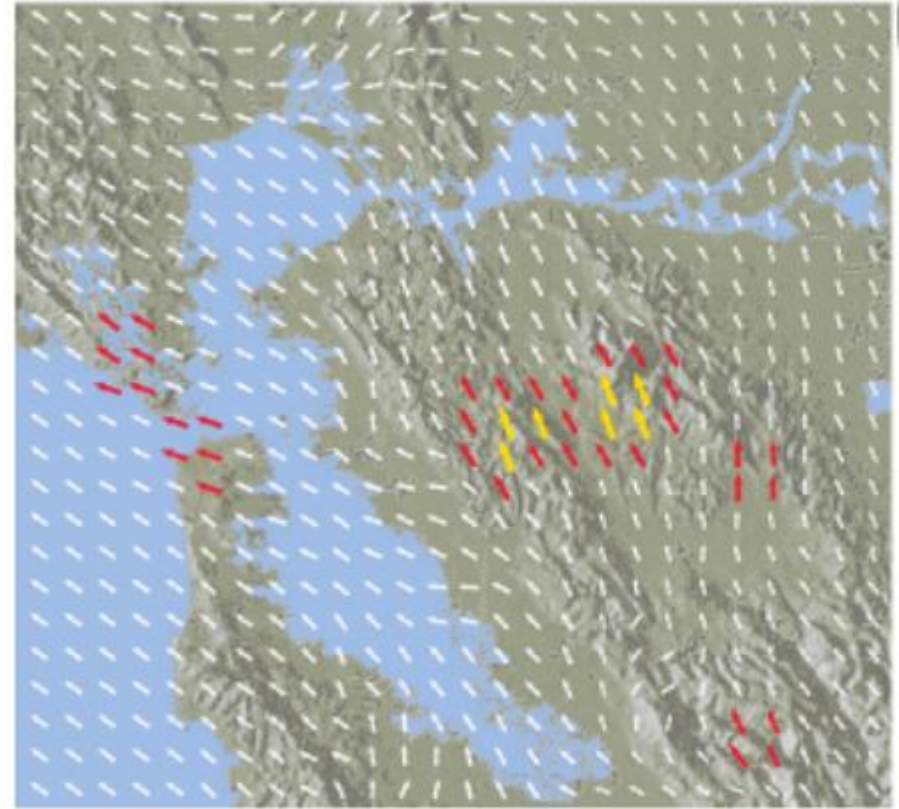


Vector Fields

Velocity vector fields showing San Francisco Bay wind patterns on a particular spring day



(a) 6:00 PM



(b) 6:00 AM

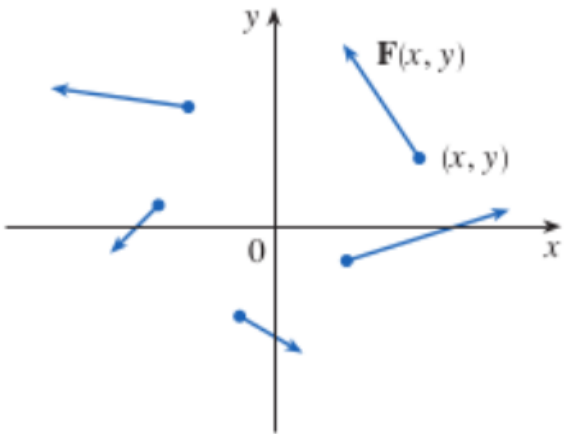
Vector Fields



Vector Fields

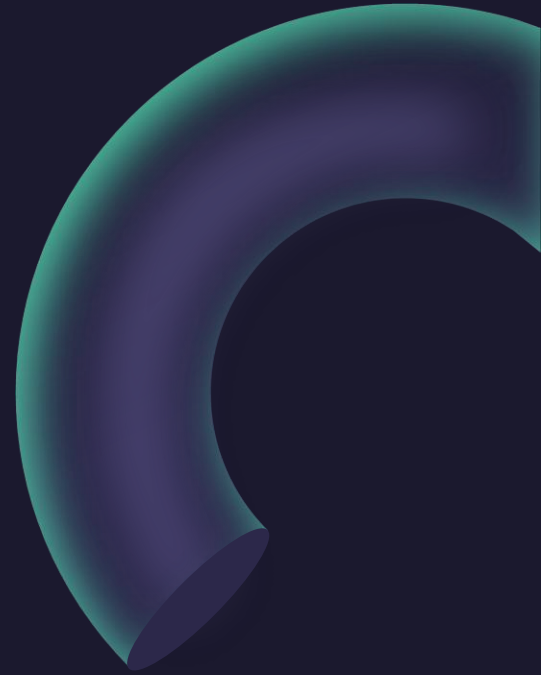
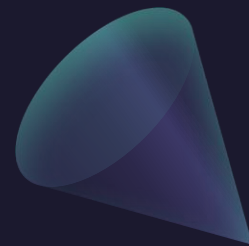
Let D be a set in $\mathbb{R}^2$ (a plane region). A **vector field on $\mathbb{R}^2$** is a function $\mathbf{F}$ that assigns to each point (x, y) in D a two-dimensional vector $\mathbf{F}(x, y)$.

Let E be a subset of $\mathbb{R}^3$. A **vector field on $\mathbb{R}^3$** is a function $\mathbf{F}$ that assigns to each point (x, y, z) in E a three-dimensional vector $\mathbf{F}(x, y, z)$.

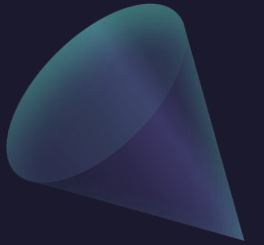


Example: Sketch a vector field on the plane given by

$$F(x, y) = (-y, x) = -y \cdot i + x \cdot j$$



Questions?



Thank you

Until next time.

